A CRM APPLICATION FOR THECHAISHOP&CO

User Story:

As a sales representative at TheChaiShop&Co, I want to efficiently manage customer relationships and streamline sales processes using our CRM application.
I want to easily access and update employer details, as well as distinguish between online and offline customer records using record types, allowing for personalized interactions.
Additionally, I need the application to automate sales calculations, such as order quantity and total bill amount, to ensure accurate and efficient transactions.
Moreover, I require the ability to generate insightful reports and dashboards to summarize and analyze sales data effectively, aiding in decision-making and strategy development.
Lastly, I would like the application to automate email notifications to customers on their birthdays, informing them of special offers tailored to celebrate their special day, enhancing customer engagement and loyalty.

1. INTRODUCTION

1.1 Project Overview :

The project “A CRM Application for TheChaiShop&Co” focuses on developing a Salesforce-based Customer Relationship Management (CRM) system to manage customer, employer, and enterprise accounts efficiently. The system automates sales processes, maintains customer records, generates reports, and improves customer engagement through automated notifications.

The project user story and objectives are defined in the initial section of the document

1.2 Purpose :

The purpose of this project is to:

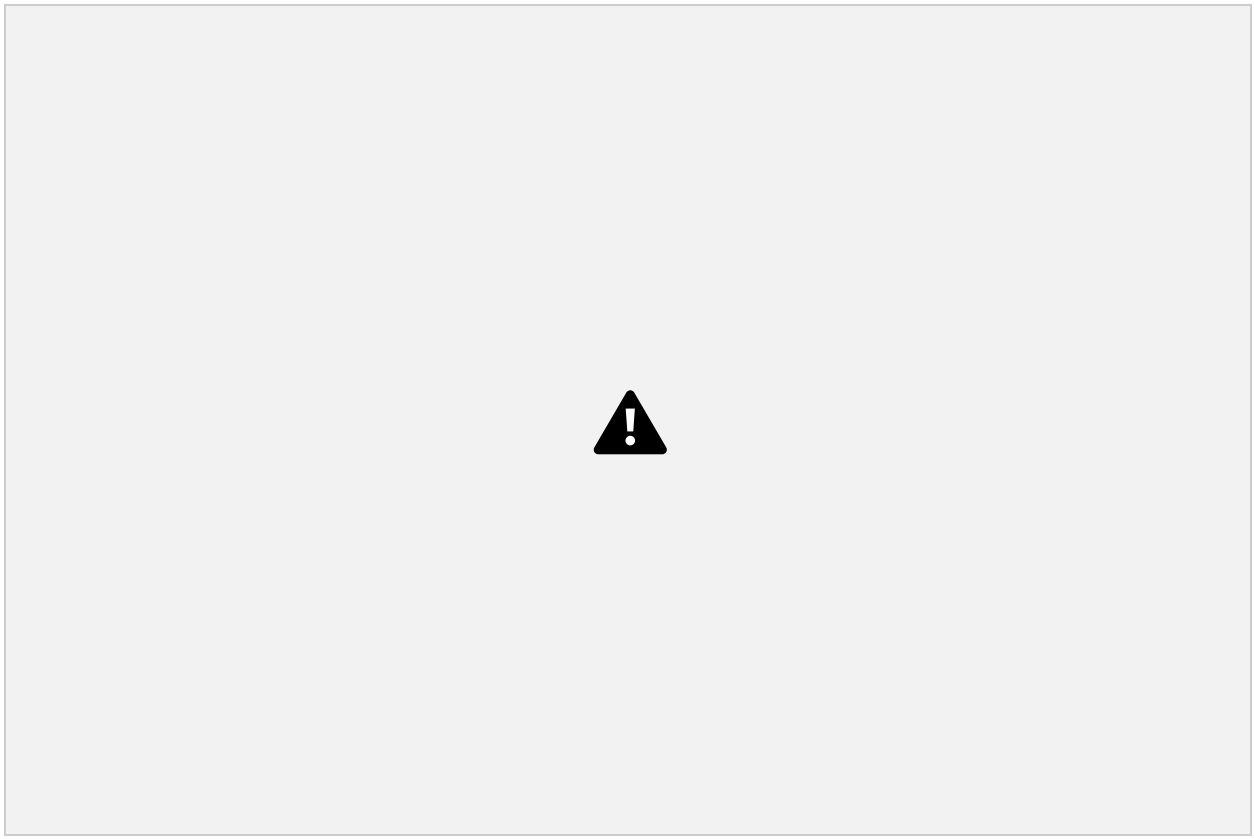
- Streamline customer relationship management

- Automate billing and order calculations
- Improve data accuracy
- Enable data-driven decision-making
- Enhance customer engagement through email automation

2. IDEATION PHASE

2.1 Problem Statement (Customer Problem Statement Template)

PS	**I am (Customer)**	**I'm trying to**	**But**	**Because**	**Which makes me feel**
PS-1	Sales Representative	Manage customer orders and billing	Calculations are often incorrect	Order quantity and billing are manual	Stressed and inefficient
PS-2	Store Manager	Monitor sales and customer records	Reports are delayed and incomplete	Data is stored in multiple files	Confused and dissatisfied
PS-3	Customer	Receive personalized offers and updates	I do not get timely notifications	There is no automated communication	Disconnected and undervalued



2.2 Empathy Map Canvas

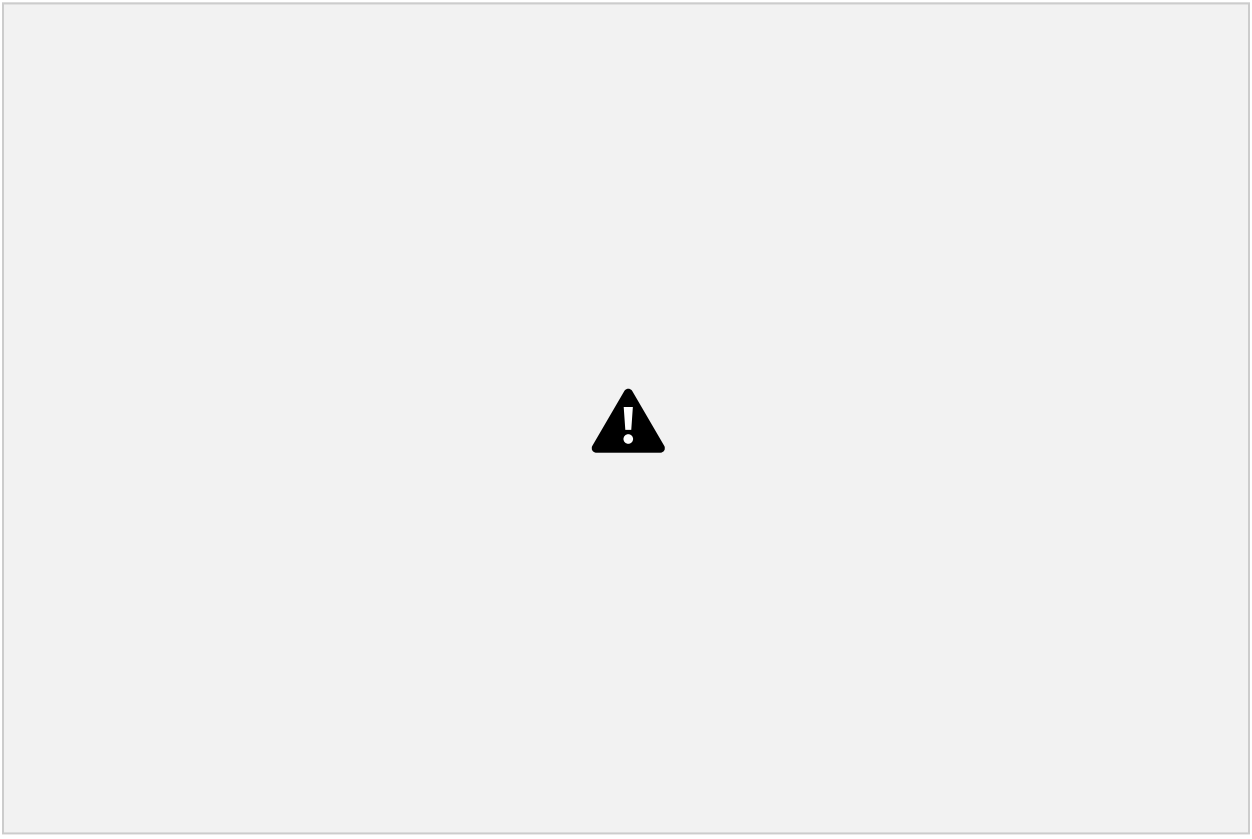
Target Users

- Sales Representatives
- Managers
- System Administrators
- Business Analysts

Empathy Analysis

Aspect	**Description**
Think & Feel	Need accurate customer data and reports
Hear	Feedback from management
See	Multiple registers and spreadsheets

Say & Do	Enter data manually
Pain	Errors, delays, data loss
Gain	Automation, fast reports



3.REQUIREMENT ANALYSIS PHASE

3.1 Customer Journey Map

Journey Stages:

- Login
- Access CRM Application
- Enter Customer Data
- Generate Orders

- Calculate Bills
- View Reports
- Decision Making

3.2 Solution Requirements

Functional Requirements

FR No	Requirement
FR-1	User Login
FR-2	Object Creation
FR-3	Data Entry
FR-4	Billing Automation
FR-5	Report Generation
FR-6	Email Automation

3.3Technology Stack

Layer	Technology
Frontend	Salesforce Lightning
Database	Salesforce Cloud
Automation	Flows
Backend	Salesforce Platform
Analytics	Reports & Dashboards

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

- The project followed an Agile-based development approach with defined milestones and sprints.
- A Salesforce Developer account was created to set up the development environment.
- Custom objects were designed to store customer, employer, and enterprise data.
- Custom tabs and a Lightning Application were developed for easy system navigation.
- Fields, relationships, and global picklists were configured to ensure data consistency.
- Page layouts and record types were created based on user roles and customer categories.
- Validation rules were implemented to prevent incorrect data entry.
- Email templates and automated flows were developed for customer notifications.
- Reports and dashboards were created for performance monitoring and analysis.
- Regular sprint reviews and testing were conducted to track progress and resolve issues.
- The project was completed within the planned timeline through systematic scheduling.

Project Development Phase:

Project Flow

Milestone 1 : Creation of developer account
Milestone 2 : Object Creation
Milestone 3 : Tabs
Milestone 4 : The Lightning App
Milestone 5 : Global Picklist
Milestone 6 : Fields and Relationships
Milestone 7 : Creation of Page Layout
Milestone 8 : Creation of Record Types
Milestone 9 : Validation rules
Milestone 10 : Email Templates
Milestone 11 : Flows

Milestone 12 : Reports and Dashboards
Milestone 13 : Conclusion

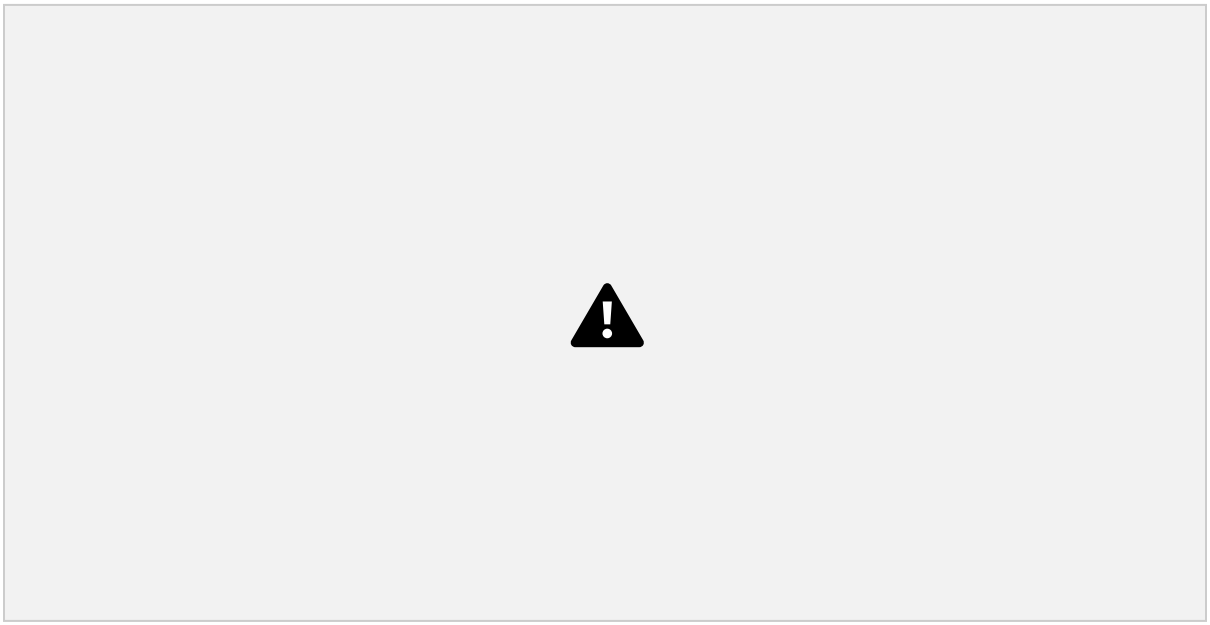
Implementation

Milestone 1 - Salesforce developer account creation :

Activity 1: Creating Developer Account:

Creating a developer org in salesforce.

1. Go to https://developer.salesforce.com/signup
2. On the sign up form, enter the following details :



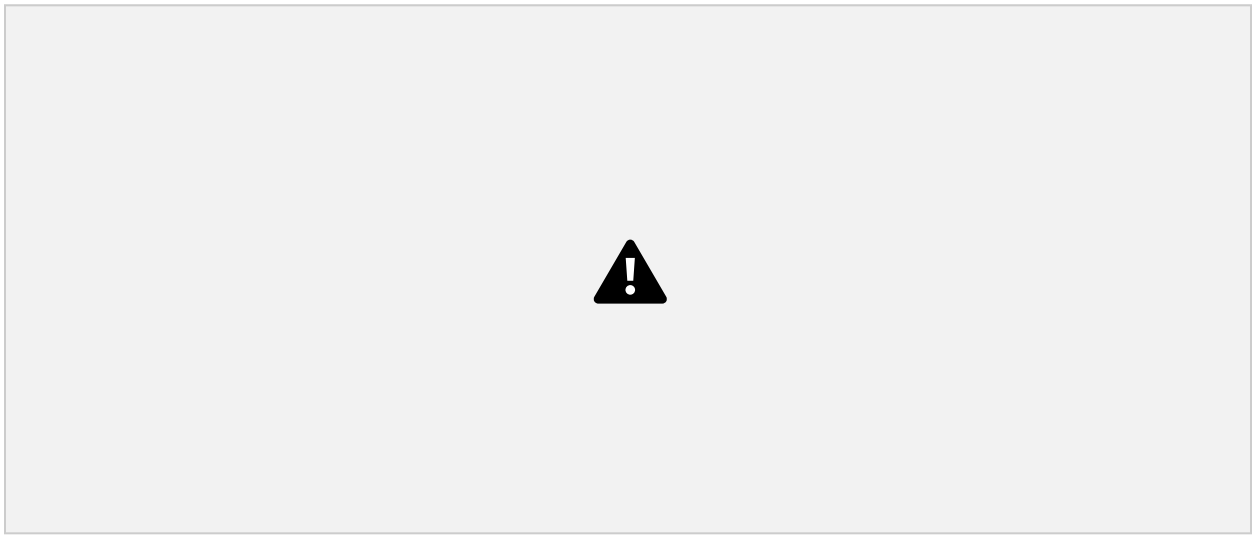
1) First name & Last name
2) Email
3) Role : Developer
4) Company : College or Company Name
5) County : India

1) Postal Code : pin code
2) Username : should be a combination of your name and company

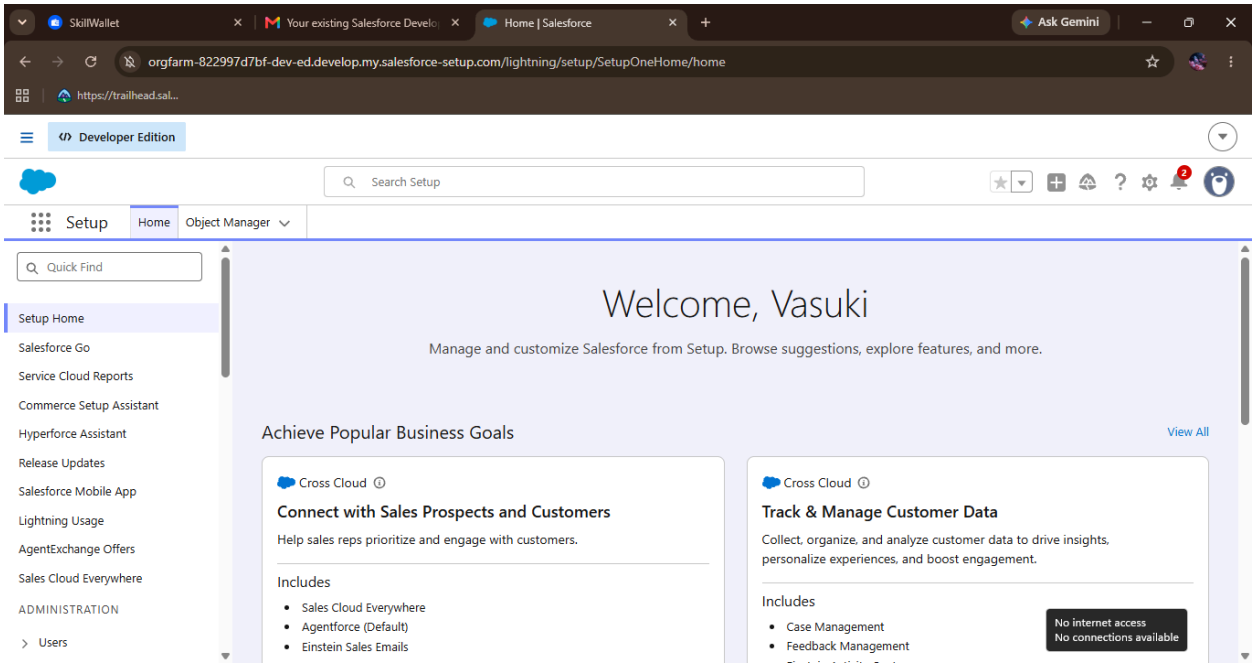
This need not be an actual email id, you can give anything in that format : username@organization.com
Click on sign me up after filling these.

Activity 2: Account Activation :

1. Go to the inbox of the email that you used while signing up. Click on the verify account to activate your account. The email may take 5-10mins.



1. Click on Verify Account.
2. Give a password and answer a security question and click on change password.



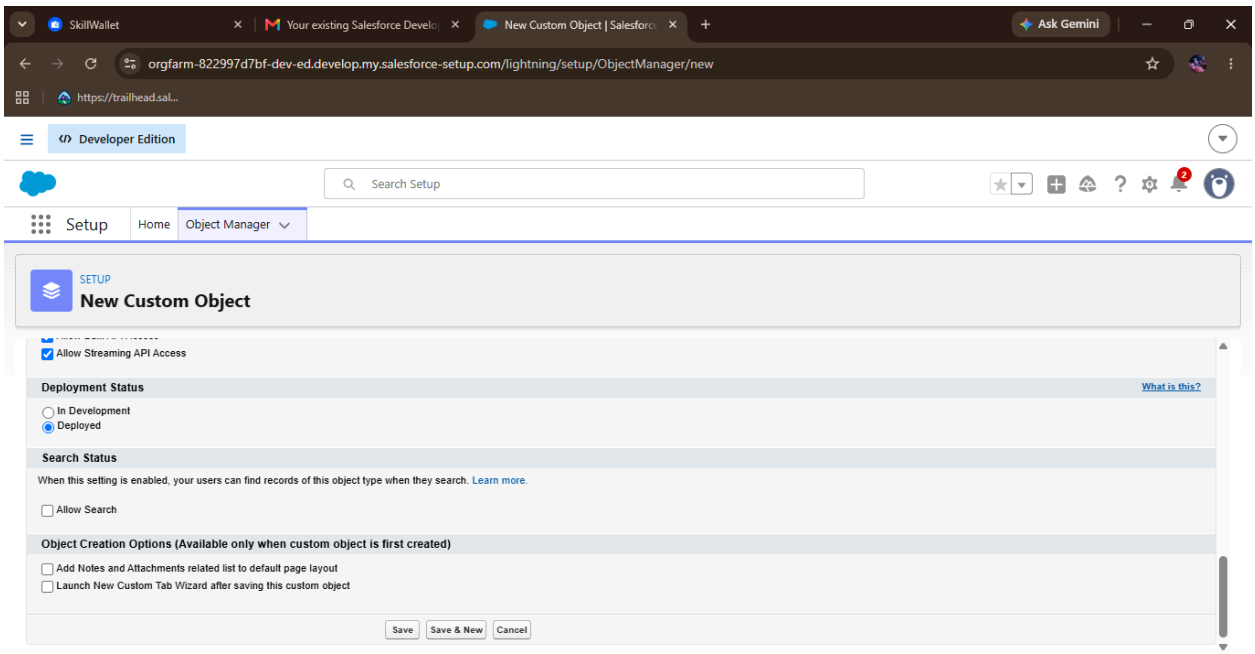
3.Then you will redirect to your salesforce setup page.

Milestone 2 - Objects

1. In your Salesforce org, click gear icon on the top left and select Setup to open Setup.

Activity 1 : Employer Account Object

1. From the setup page → Click on Object Manager → Click on Create → Click on Custom Object.
    - **Label** : Employer Account
    - **Plural Label** : Employer Accounts
    - Enter Record Name Label and Format
        Record Name : Employer Account Name
        Data Type **:** Text
    - **In Optional features :** check the boxes for Allow Reports | Allow Activities | Track Field History.
    - **Search Status :** check the box for Allow Search.
    - Save



Activity 2 : Customer Account Object

1. From the setup page → Click on Object Manager → Click on Create → Click on Custom Object.
    - **Label** : Customer Account
    - **Plural Label** : Customer Accounts
    - Enter Record Name Label and Format
        Record Name : Customer Account Name
        Data Type **:** Text
    - **In Optional features :** check the boxes for Allow Reports | Allow Activities | Track Field History.
    - **Search Status :** check the box for Allow Search.
    - Save

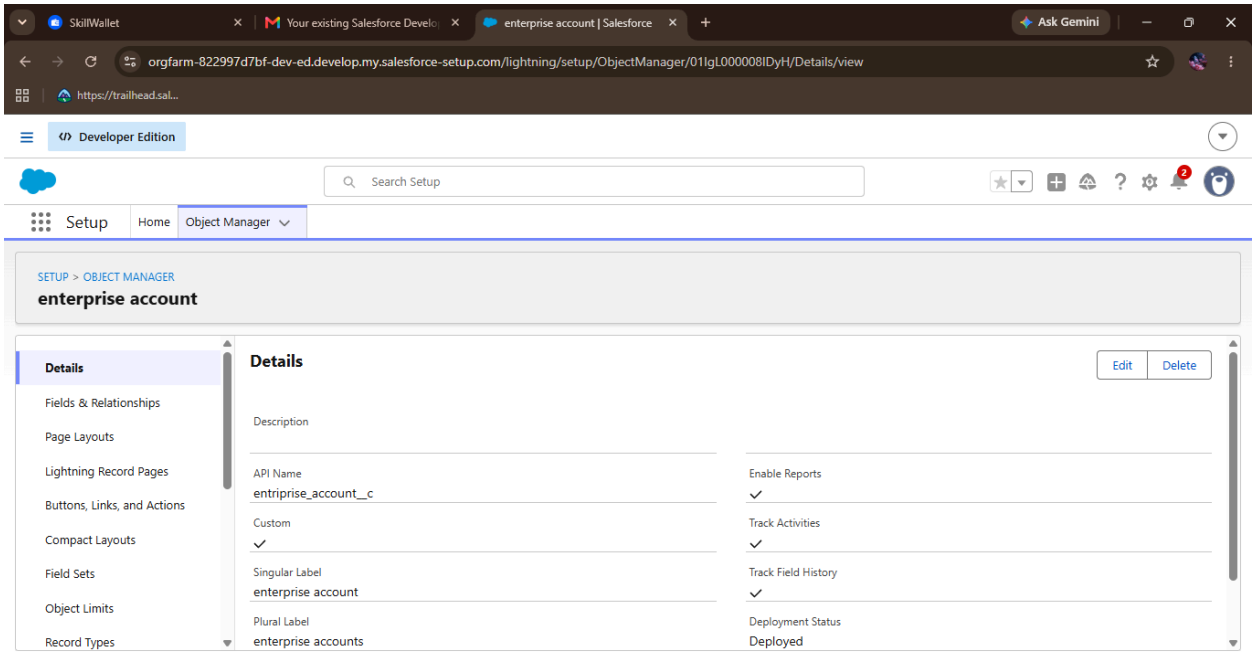
Activity 3 : Enterprise Account Object

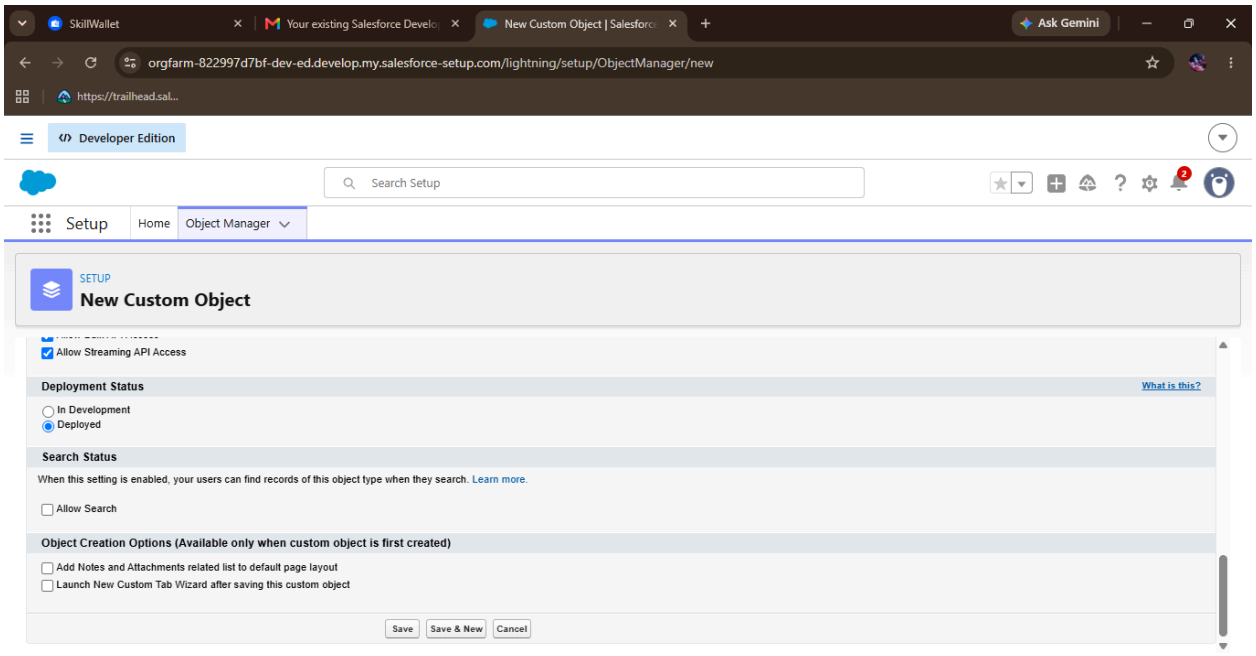
1. From the setup page → Click on Object Manager → Click on Create → Click on Custom Object.
    - **Label** : Enterprise Account
    - **Plural Label** : Enterprise Accounts
    - Enter Record Name Label and Format
        Record Name : Enterprise Account Name
        Data Type **:** Text
    - **In Optional features :** check the boxes for Allow Reports | Allow Activities | Track Field History.
    - **Search Status :** check the box for Allow Search.
    - Save

Milestone 3 - Tabs

Activity 1 : Custom tabs creation

1. Go to setup page → type Tabs in Quick Find bar → click on tabs → New (under custom object tab)

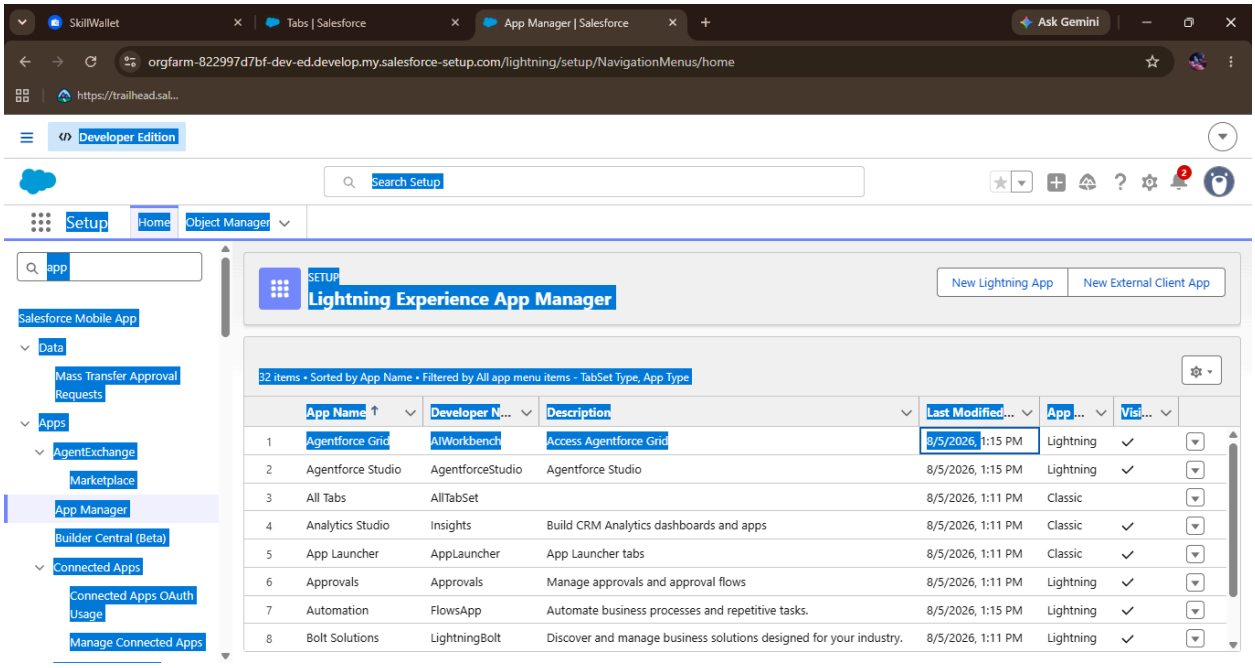




Milestone 4 - The Lightning App

Activity 1 : Create a lightning app

1. Go to setup page → search “app manager” in quick find → select “app manager” → click on New lightning App.



2. Fill the app name in app details and branding as follow
   App Name : TheChaiShop&Co
   Developer Name : Auto Populates
   Add image optional (if you want to give any image you can otherwise not mandatory) - Add Primary color Hex or leave it to default.
   Then click Next.

3. In App options
   Navigation style : Standard navigation
   Setup experience : setup
   Supporters form factors : Desktop and phone
   Then click Next

4. In Utility items
   Utility Bar alignment : Default
   Then click Next.

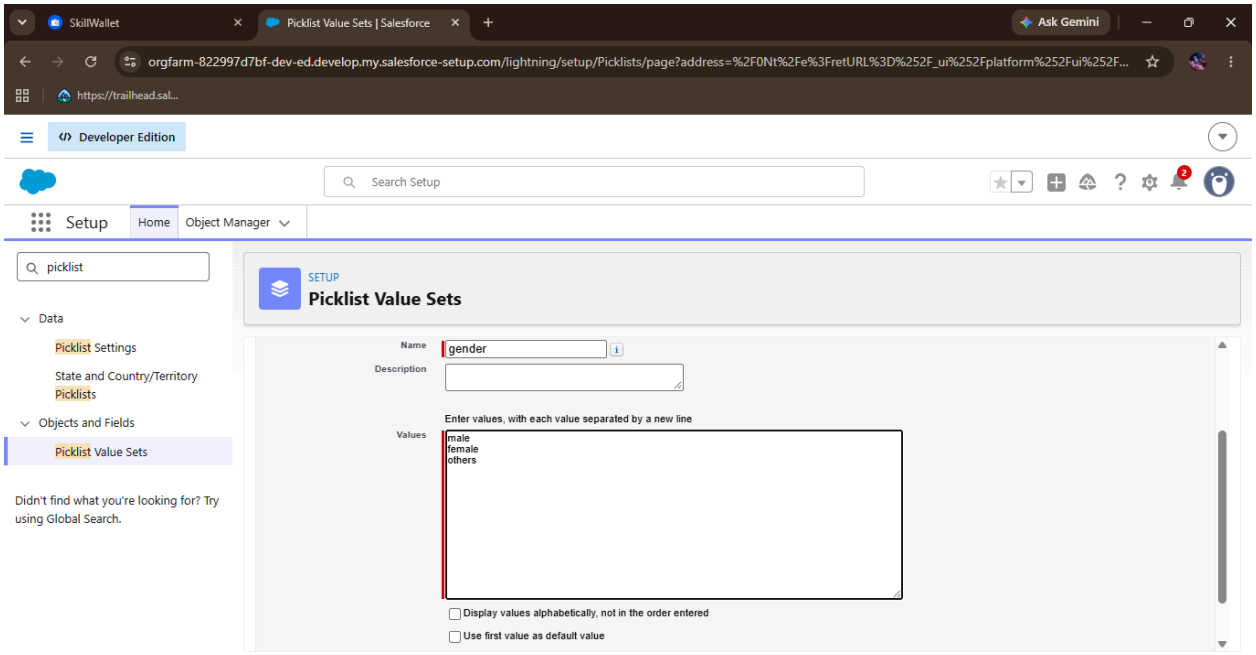
5. Navigation items
   Select the created Custom Objects and required standard objects
   - Customer Accounts
     Employer Accounts
     Enterprise Accounts

Reports
Dashboards

Milestone 5 : Global Picklist

Activity 1 : Create Global PIcklist

1. In Quick find box search for : Picklist value sets and click New.



1. Create three more Global picklists following the same steps.
    - **Label** : Items
      Name : Auto populates
      Values : Platinum Tea Powder
      Double Delight Tea Powder
      Super Dust Tea Powder
      Osmania Biscuits (Value Pack)
      Osmania Biscuits (Premium Pack)
      Dry Fruit Biscuits
      Fruit Biscuits
      Salt Biscuits
      Multigrain Biscuits
      Ragi Biscuits
      Double Choco Chip Cookies

Butter Almond Cookies
Melting Moments Cookies
Honey Oats Cookies
Delight Box
Signature Box
Assorted Box

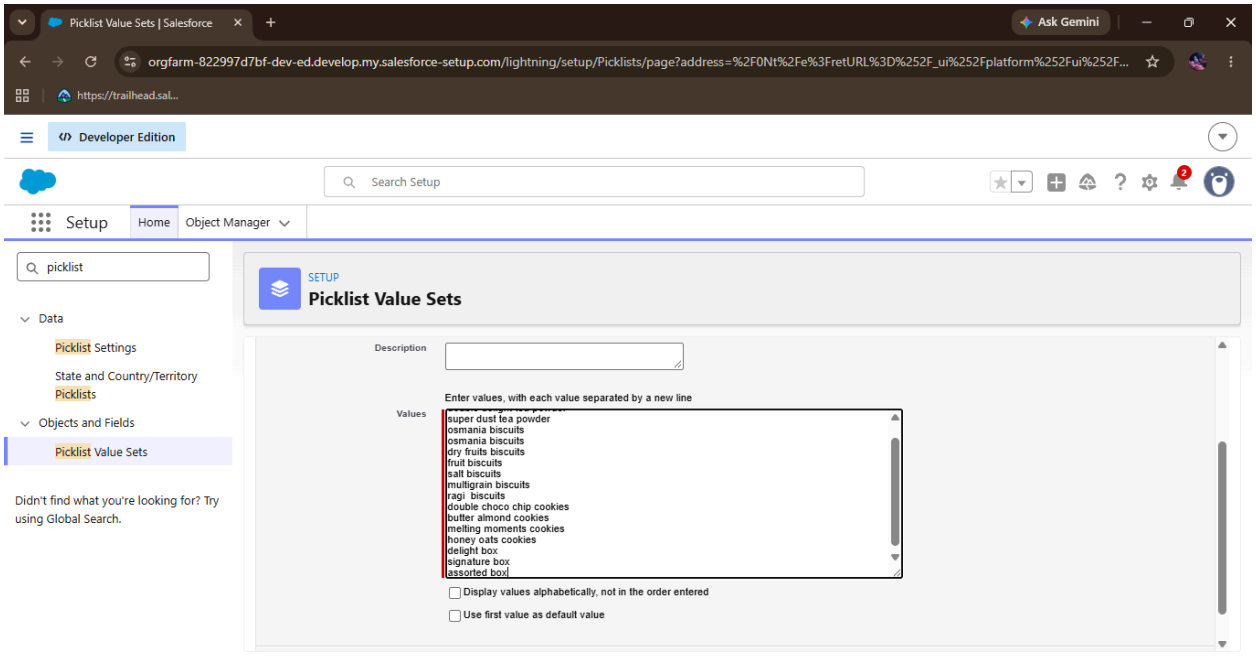
Click **Save & New.**

- **Label** : Payment Mode
  Name : Auto populates
  Values : Gpay
  PhonePe
  Paytm
  Credit or Debit card
  Cash.

  Click **Save & New.**

- **Label :** Type of Items
  Name : Auto populates
  Values : Tea powder
  Cookies
  Gift Hampers

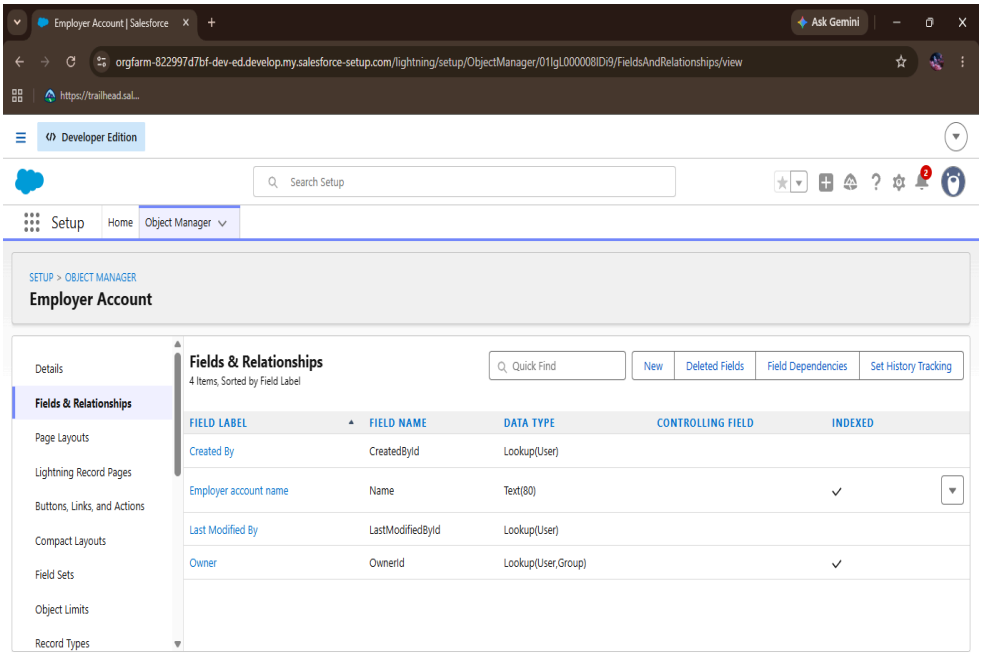
  Click **Save.**



Milestone 6 : Fields & Relationships

Activity 1 : Create Custom fields for Employer Account

1. In your Salesforce org, click gear icon on the top left and select Setup to open Setup.
2. Click the Object Manager tab. If you don't see it, enter Object Manager in the Quick Find box.
3. From the object manager page, In the Quick Find box, Search for the custom object you just created : **Employer Account**
4. From the sidebar, click Fields & Relationships. Notice that there are already some fields there. Those are the standard fields.
5. Click New to create a custom field. Tip: Before creating a new field, do a quick search to make sure a similar one doesn't already exist.



6. Next, choose a data type as Text. Choosing a data type helps you format the field input.

Activity 2: Create Custom fields for Customer Account

1. Data Type : Text
   Field Label: First Name

Field Name : Auto Populates
Length : 25
Required : Check the box
Click on Next → Next → Save and new.

2. Data Type : Text
   Field Label: Last Name
   Field Name : Auto Populates
   Length : 25
   Required : Check the box
   Click on Next → Next → Save and new.

3. Data Type : **PickList**
   Field Label: Gender
   Values : Use **Global Picklist** value set and select Gender from the dropdown menu.
   Field Name : Auto Populates.
   Required : Check the box
   Click on Next → Next → Save and new.

4. Data Type : Phone
   Field Label : Phone Number
   Field Name : Auto Populates
   Required : Check the box
   Click on Next → Next → Save and new.

5. Data Type : Date
   Field Label : Date of Birth
   Field Name : Auto Populates
   Required : check the box
   Click on Next → Next → Save and new.

6. Data Type : **Formula**
   Field Label : Date of Birthday is today ?
   Field Name : Auto Populates
   Formula Return : **Checkbox**
   Formula :

```
IF(
AND(
DAY( TODAY() ) = DAY( Date_of_Birth__c ) ,
MONTH ( TODAY()) = MONTH( Date_of_Birth__c )),
```

TRUE,
FALSE)

Click on Next → Next → Save and new.

7. Data Type : Email
   Field Label : Email ID
   Field Name : Auto Populates
   Required : Check the box
   Click on Next → Next → Save and new.

8. Date Type : Text
   Field Label : Address
   Field Name : Auto Populates
   Length : 125
   Click on Next → Next → Save and new.

9. Date Type : **Multi Select Picklist**
   Field Label : Type of Items
   Values : Use **Global Picklist** value set and
   select **Type of Items** from the dropdown menu.
   Field Name : Auto Populates.
   Required : Check the box
   Click on Next → Next → Save and new.

10. Date Type : **Multi Select Picklist**
    Field Label : Products purchased
    Values : Use **Global Picklist** value set and
    select **Items** from the dropdown menu.
    Field Name : Auto Populates.
    Required : Check the box
    Click on Next → Next → Save and new.

11. Date Type : **Multi Select Picklist**
    Field Label : Chai
    Value : Enter value, with each value separated by a new line.
    TheChaiShop&Co Special Tea
    Single Tea
    Masala Tea
    Lemon Tea
    Ginger Tea
    Green Tea

Sugar Free Tea
Click on Next → Next → Save and new.

12. Data Type : **Picklist**
    Field Label : Payment Mode
    Values : Use **Global Picklist** value set and
    select **Payment Mode** from the dropdown menu.
    Field Name : Auto Populates.
    Required : Check the box
    Click on Next → Next → Save and new.

13. Data Type : **Lookup Relationship**
    Related To : Employer Account
    Field Label : Associated Employer
    Field Name : Auto Populates
    Click on Next → Next → Save and new.

Activity 3: Create Custom fields for Enterprise Account

1. Data Type : Text
   Field Label: Enterprise Name
   Field Name : Auto Populates
   Length : 25
   Required : Check the box
   Click on Next → Next → Save and new.

2. Data Type : Phone
   Field Label : Phone Number
   Field Name : Auto Populates
   Required : Check the box
   Click on Next → Next → Save and new.

3. Data Type : Email
   Field Label : Email ID
   Field Name : Auto Populates
   Required : Check the box
   Click on Next → Next → Save and new.

4. Date Type : Text
   Field Label : Address

Field Name : Auto Populates
Length : 125
Click on Next → Next → Save and new.

5. Date Type : **Multi Select Picklist**
   Field Label : Type of Items
   Values : Use **Global Picklist** value set and select **Type of Items** from the dropdown menu.
   Field Name : Auto Populates.
   Required : Check the box
   Click on Next → Next → Save and new.

6. Date Type : **Multi Select Picklist**
   Field Label : Products purchased
   Values : Use **Global Picklist** value set and select **Items** from the dropdown menu.
   Field Name : Auto Populates.
   Required : Check the box
   Click on Next → Next → Save and new.

7. Data Type : **Lookup Relationship**
   Related To : Employer Account
   Field Label : Associated Employer
   Field Name : Auto Populates
   Click on Next → Next → Save and new.

8. Data Type : **Formula**
   Field Label : Order Quantity
   Field Name : Auto Populates
   Formula Return : **Number**
   Enter Formula = Quantity (Number) =

```
IF(INCLUDES( Products_Purchased__c , "Platinum Tea Powder"), 1, 0 ) +
        IF(INCLUDES( Products_Purchased__c , "Double Delight Tea Powder"), 1, 0 ) +
        IF( INCLUDES( Products_Purchased__c , "Super Dust Tea Powder") , 1, 0) +
        IF(INCLUDES( Products_Purchased__c , "Osmania Biscuits (Value Pack)"), 1, 0 ) +
        IF(INCLUDES( Products_Purchased__c , "Osmania Biscuits (Premium Pack)"), 1, 0 ) +
        IF( INCLUDES( Products_Purchased__c , "Dry Fruit Biscuits") , 1, 0) +
        IF(INCLUDES( Products_Purchased__c , "Fruit Biscuits"), 1, 0 ) +
        IF(INCLUDES( Products_Purchased__c , "Salt Biscuits"), 1, 0 ) +
        IF(INCLUDES( Products_Purchased__c , "Multigrain Biscuits"), 1, 0 ) +
        IF(INCLUDES( Products_Purchased__c , "Ragi Biscuits"), 1, 0 ) +
        IF( INCLUDES( Products_Purchased__c ,"Double Choco chip Cookies") , 1, 0) +
        IF(INCLUDES( Products_Purchased__c , "Butter Almond Cookies"), 1, 0 ) +
        IF(INCLUDES( Products_Purchased__c , "Melting Moments Cookies"), 1, 0 ) +
        IF( INCLUDES( Products_Purchased__c , "Honey Oats Cookies") , 1, 0) +
```

```
IF(INCLUDES( Products_Purchased__c , "Delight Box"), 1, 0 ) +
IF(INCLUDES( Products_Purchased__c , "Signature Box"), 1, 0 ) +
IF( INCLUDES( Products_Purchased__c , "Assorted Pack") , 1, 0)
```

Click on Next → Next → Save and new.

9. Data Type : **Formula**
   Field Label : Total Bill
   Field Name : Auto Populates
   Formula Return : **Number**
   Enter Formula = Quantity (Number) =

```
IF(INCLUDES( Products_Purchased__c , "Platinum Tea Powder"), 175, 0 ) +
IF(INCLUDES( Products_Purchased__c , "Double Delight Tea Powder"), 150, 0 ) +
IF( INCLUDES( Products_Purchased__c , "Super Dust Tea Powder") , 125, 0) +
IF(INCLUDES( Products_Purchased__c , "Osmania Biscuits (Value Pack)"), 50, 0 ) +
IF(INCLUDES( Products_Purchased__c , "Osmania Biscuits (Premium Pack)"), 80, 0 ) +
IF( INCLUDES( Products_Purchased__c , "Dry Fruit Biscuits") , 110, 0) +
IF(INCLUDES( Products_Purchased__c , "Fruit Biscuits"), 160, 0 ) +
IF(INCLUDES( Products_Purchased__c , "Salt Biscuits"), 110, 0 ) +
IF(INCLUDES( Products_Purchased__c , "Multigrain Biscuits"), 170, 0 ) +
IF(INCLUDES( Products_Purchased__c , "Ragi Biscuits"), 190, 0 ) +
IF( INCLUDES( Products_Purchased__c ,"Double Choco chip Cookies") , 200, 0) +
IF(INCLUDES( Products_Purchased__c , "Butter Almond Cookies"), 110, 0 ) +
IF(INCLUDES( Products_Purchased__c , "Melting Moments Cookies"), 150, 0 ) +
IF( INCLUDES( Products_Purchased__c , "Honey Oats Cookies") , 160, 0) +
IF(INCLUDES( Products_Purchased__c , "Delight Box"), 850, 0 ) +
IF(INCLUDES( Products_Purchased__c , "Signature Box"), 1500, 0 ) +
IF( INCLUDES( Products_Purchased__c , "Assorted Pack") , 600, 0)
```

10. Data Type : **Picklist**
    Field Label : Payment Mode
    Values : Use **Global Picklist** value set and
    select **Payment Mode** from the dropdown menu.
    Field Name : Auto Populates.
    Required : Check the box
    Click on Next → Next → Save.

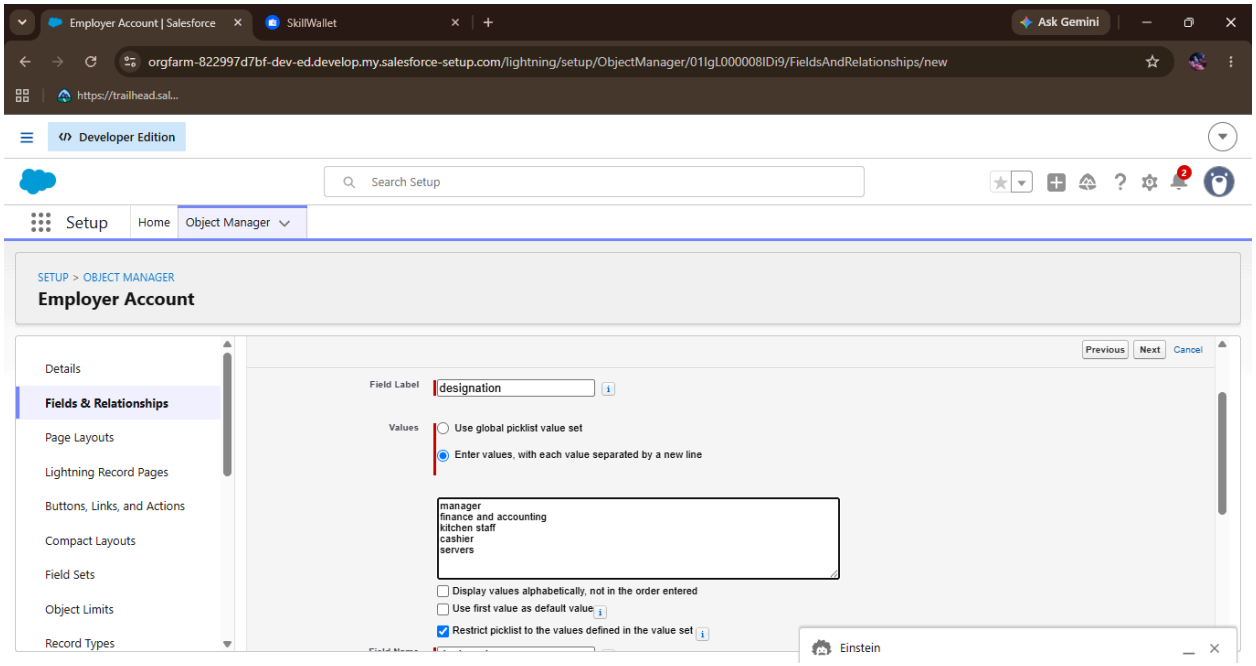
Milestone 7 : Page Layouts

For the personalized field view in Customer Accounts, let's create two page layouts Online orders and Walk-In orders.

Activity 1: Online Orders page layout

1. Go to the setup page → click on object manager → From drop down click edit for **Customer Account** object.

2. Click on the Page Layouts → click New.



Milestone 8 - Record Types

To Differentiate the records and data stored in Customer Accounts, let's create two different Record Types.
Online orders and Walk-In orders.

Activity 1: Online Orders Record Type

1. Go to the setup page → click on object manager → From drop down click edit for **Customer Account** object.

2. Click on the Record Types → click New.

3. Enter Details :
   Existing Record Type : Master
   Record Type Label : Online Order

Activity 2: Walk-In Orders Record Type

1. Go to the setup page → click on object manager → From drop down click edit for **Customer Account** object.

2. Click on the Record Types → click New.

3. Enter Details :
   Existing Record Type : Master
   Record Type Label : Walk-In Order
   Record Type Name : Auto Populates
   Active : Tick the checkbox
   Click Next

4. In Assign Page Layouts
   - Select Apply one layout to all profiles : select **Walk-In orders** from the drop down menu.

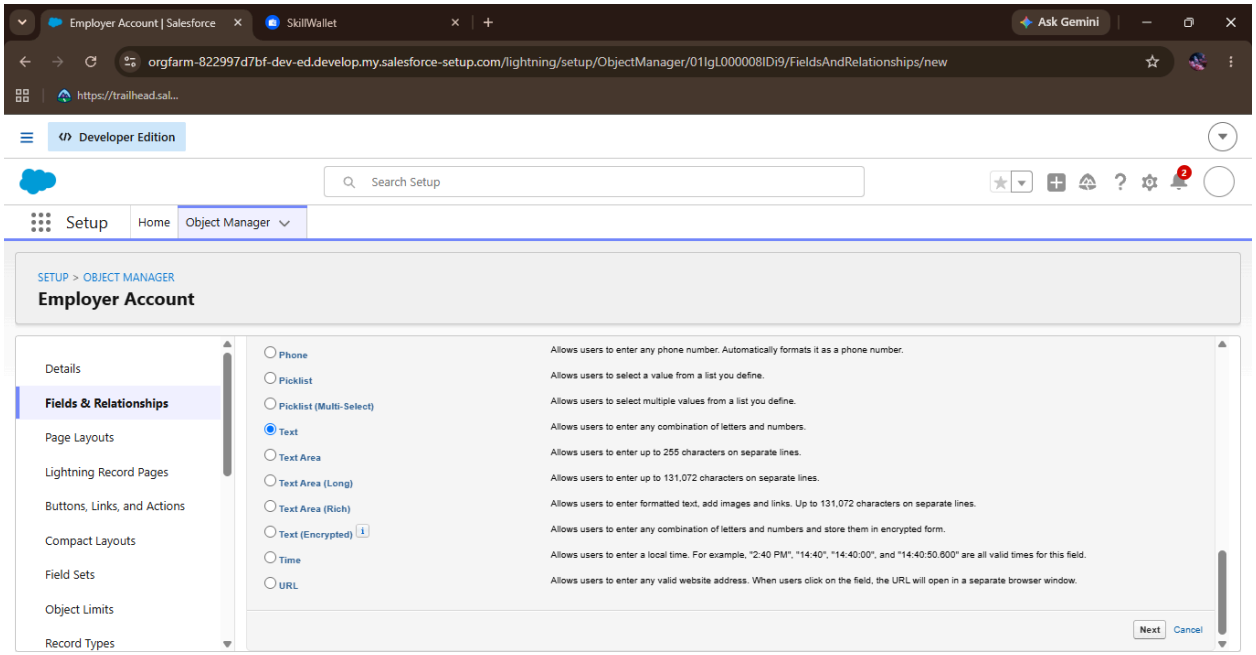
5. Click **Save.**

Milestone 9 - Validation Rules

Activity 1: On Employer Account

1. Go to the setup page → click on object manager → From drop down click edit for **Employer Account** object.

2. Click on the validation rule → click New.



Activity 2: On Customer Account

1. Go to the setup page → click on object manager → From drop down click edit for **Customer Account** object.

2. Click on the validation rule → click New.

3. **Enter Rule Name** : Phone Number

   Error Condition Formula :

```
AND(
NOT(ISBLANK( Phone_Number__c )),
NOT(REGEX( Phone_Number__c , "^[1-9]{10}$")),
NOT(REGEX( Phone_Number__c , "^\\d+$"))
)
```

   Error Message :
   The phone number field must not be blank.

Activity 3: On Enterprise Account

1. Go to the setup page → click on object manager → From drop down click edit for **Customer Account** object.

2. Click on the validation rule → click New.

3. **Enter Rule Name** : Phone Number

   Error Condition Formula :

```
AND(
NOT(ISBLANK( Phone_Number__c )),
NOT(REGEX( Phone_Number__c , "^[1-9]{10}$")),
NOT(REGEX( Phone_Number__c , "^\\d+$"))
)
```

   Error Message :
   The phone number field must not be blank.
   The phone number must consist of exactly 10 digits.
   The phone number must contain only numeric digits
   (no spaces, dashes, parentheses, etc.)

4. Click **Save.**

Milestone 10 - Email Template

Activity 1: Email Template

1. Go to setup, In quick find box enter Email template → click on classic Email Template.

2. Click on→ New Email Template

3. Choose the type of email template you would like to create : **Text** and click **Next**.

Milestone 11 - Flows

Let’s create a scheduled Triggered flow to automate the Birthday offer mails.

Activity 1: Schedule Triggered Flow

Milestone 12 - Reports & Dashboards

Activity 1: Customer Account Sales Report

1. Click App Launcher and select TheChaiShop&Co.
2. Click Reports
3. Click New Report

Milestone 12 - Conclusion

In conclusion, the development of TheChaiShop&Co CRM application marks a significant stride in enhancing the store's operational efficiency and customer relationship management capabilities. Seamlessly integrating employer details, customer information (both online and offline), and enterprise data, this application streamlines business processes and enables informed decision-making. The automation of sales calculations, including order quantity and total bill amount, reduces errors and empowers employees to focus on delivering exceptional service. Moreover, automated email notifications for customers' birthdays foster personalized communication and enhance customer engagement.

7. FUNCTIONAL & PERFORMANCE TESTING

During the testing phase, screenshots were captured from the Salesforce Lightning interface to verify correct system behavior. These include:

- Record creation for Employer, Customer, and Enterprise objects
- Validation rule testing for Aadhar, PAN, and phone numbers
- Verification of record types and page layouts
- Formula field validation for order quantity and billing
- Flow execution for birthday email alerts
- Report and dashboard generation

Performance testing ensured fast response time and system stability during concurrent usage.

8. ADVANTAGES & DISADVANTAGES

Advantages

- Centralized data management
- Reduced manual work
- Automated calculations
- Improved customer engagement
- Real-time analytics

Disadvantages

- Requires Salesforce license
- Internet dependency
- Limited offline access

9. Conclusion:

The CRM Application for TheChaiShop&Co successfully improves business operations by integrating customer data, automating billing, and enabling intelligent reporting. The system reduces errors, saves time, and enhances customer relationships through automation and analytics.

10. FUTURE SCOPE

- Mobile CRM integration
- AI-based sales prediction
- WhatsApp/SMS alerts
- Inventory management
- Multi-branch support
- Advanced customer profiling

PROJECT REPORT ON A CRM APPLICATION FOR THE CHAI SHOP & CO

SUBMITTED TO : KAMBAN COLLEGE OF ARTS AND SCIENCE FOR WOMEN

DEPARTMENT OF DATA SCIENCE

ACADEMIC YEAR:2024-2027

PROJECT NAME DETAILS

TEAM LEAD: VASUKI P (20624U48050) vasukiparamasivam821@gmail.com

TEAM MEMBERS:VAISHNAVI.R vaishnaviragupathi.17@gmail.com

USHARANI.R usharanisaravanan007@gmail.com